Gap Analysis – Lab 1 baseline vs. Mini-Project Part-1

Baseline files (kept unchanged, now stored in the repository's `Lab-1/` folder): `Requirements_Table.docx`, `UseCase_Flow_Specification.docx`, `UML_UseCase_Diagram.pdf`. No source code for the portal exists yet (Part-1 is documentation only; the other folders in the repository are separate course labs).

1. EXISTING

Item	Content
Requirements	FR-001…FR-005 and NFR-001, NFR-002 with type, priority, acceptance criteria, rationale (Lab 1 mandated exactly these 5 + 2).
Use case flow	UC-03 Reserve Equipment Slot: preconditions, postconditions, 8-step main flow, alternate flows A1 (calibration) and A2 (slot conflict).
UML use-case diagram	Actors Student, Lab Technician, Notification System; 12 use cases; 3 «include» from Reserve Equipment Slot, 1 «extend» (Apply Late-Return Penalty → Return Equipment).

2. MISSING (required by the Part-1 assignment)

Assignment item	Status before
IEEE-style SRS with introduction, FR, NFR, interfaces	missing → `SRS.md`
Security objectives (≥ 2) and requirements (≥ 2)	missing → SRS §4 (4 objectives, 7 requirements)
IEEE-style Test Plan, sections 3, 4, 5, §5.1 Security Validation, traceability	missing → `Test_Plan.md`
≥ 7–10 test cases covering FR and NFR	missing → 25 test cases
Architecture: component diagram + description, pattern, traceability, security architecture	missing → Architecture §2–§7, Figure 2
Design: ≥ 2 sequence diagrams, API design, error handling	missing → Design §8–§11, Figures 3–4
IDs for use cases other than UC-03; use-case descriptions for the rest	missing → UC-01…UC-12 assigned and described

3. NEEDS REVISION

#	Existing material	Problem	Resolution (intent preserved)
1	Diagram vs. FR-004	Diagram links **Student** to *Return Equipment*, but FR-004 says a Lab Technician marks the return.	Association removed in the redrawn diagram; documented in SRS §5.1.
2	Diagram vs. requirements	*Login*, *View Reservation History*, *Update Calibration Status* and *Send Confirmation Notification* have no requirement (NFR-002 mentions calibration update only as a restriction).	Added FR-006, FR-007, FR-008, FR-009 (marked [NEW]).
3	UC-03 precondition "no conflicting reservation for the same time slot"	Not backed by a requirement.	Added FR-010.
4	FR-001	"real-time availability" is not measurable; one sentence bundles viewing, reserving, limits, token.	Kept ID and intent; reworded: availability reflects all Confirmed reservations; limits stated as 120 min / 7 days / overlap; marked [REFINED].
5	NFR-001	Bundles latency and lock isolation; "peak load" and "security standards" undefined → untestable as written.	Latency kept as NFR-001 (≤ 200 ms, load = test parameter A-03); isolation outcome split into NFR-003; "security standards" dropped (covered by SEC-REQs).
6	NFR-002	Fine as a requirement but combines RBAC and session expiry.	Kept unchanged; derived SEC-REQ-01 and SEC-REQ-02 so each half has its own test.
7	FR-003	"up to 1 hour before" ambiguous at exactly 60 min.	Assumption A-04 (inclusive), boundary test TC-07.
8	FR-004	Which reservation states can be returned is unstated.	Only Confirmed (Design §9, 409 `INVALID_RESERVATION_STATE`).
9	UC-03	Secondary actor "Lab Technician" has no step in the main flow; flow ends at A2 with no invalid-session/input, student-overlap or notification-failure paths.	Technician listing kept (fidelity) and footnoted; A3, A4 and E1 added and marked as Part-1 additions.
10	UC-03 step 5/7	The overlap check and the lock are separate steps.	Design performs them atomically in one transaction (DD-02) – same observable behaviour, needed for NFR-003.
11	Diagram	No use-case IDs.	UC-01…UC-12 assigned (UC-03 retained from the Lab 1 specification).

4. Information that Lab 1 does not provide (assumptions A-01…A-12 in SRS §2.6)

Technology stack; definition of "peak load"; slot granularity / minimum duration; behaviour on notification failure; deletion of equipment with future reservations; whether the due date changes status automatically; deployment security (HTTPS). Each was resolved with a labelled assumption – none is stated as fact – and should be confirmed with the instructor.

5. Final audit (strict-evaluator pass) – problems found and what was changed

#	Finding	Change
1	FR-001 and FR-002 bundled several verifiable statements in one sentence (not atomic).	Kept the IDs; rewrote them as clauses (a)–(e) and (a)–(c) with a per-clause pass criterion; tests now name the clause they cover.

#	Finding	Change
2	New requirements FR-006…FR-010, NFR-003 did not state where they came from.	Added a *Status / basis* column naming the Lab 1 source (diagram use case, UC-03 step/precondition, or the split of NFR-001).
3	Several security requirements were not in Lab 1 and looked invented.	Added a *Basis* column; marked SEC-REQ-03, 05, 06, 07 as *Proposed* with the reason; kept only requirements tied to a Lab 1 use case or NFR-002.
4	SEC-REQ-04 combined three obligations.	Split: SEC-REQ-04 (401 without a session) and SEC-REQ-07 (generic login error, no plain-text credentials); tests and matrix updated.
5	FR-010 read the UC-03 precondition one specific way without saying so.	Added assumption A-14 and the condition under which FR-010/TC-15 would be dropped.
6	"Relational database" was stated as fact; Lab 1 only says "database-level lock isolation".	Reworded to "database with transactions and locking (relational assumed – A-13)".
7	Documents claimed to follow IEEE Std 830/829/1016 formally.	Replaced by "IEEE-style structure as requested by the assignment; no formal compliance claimed".
8	Coverage gaps: availability display / filters (FR-001(a)) and equipment removal with future reservations (A-07) had no dedicated test; test data lacked the accounts needed by TC-16/17 and equipment EQ-05 for TC-05.	Added TC-24, TC-25 (suite is now 25); extended test data (§7).
9	API section did not say why an API exists or which endpoints are additions.	§9 now cites NFR-002 ("endpoints") and marks endpoints 10 and 12 as the only design additions.
10	Component diagram: the *ILog* arrow from the services package suggested all four services log, while the table says three.	Caption and §4.2 now state exactly which components use each interface.
11	Diagram text was too small when printed (sequence diagram with 8 participants on a portrait page).	UC-03 split into two readable parts (SD-01 part 1 / part 2); all figures are on landscape pages with larger fonts.
12	Authorization-path arrow crossed another arrow in the component diagram.	Re-routed.
13	PDF/DOCX defects found on a page-by-page read: empty shaded first row in the document-info tables, the SRS security table rendered as raw text, headings stranded before landscape figures, figures printed small, a stray control character before a figure.	Fixed in the SRS table markup and in `build_docs.py`; headings now sit on the same landscape page as their figure; PDFs and DOCX rebuilt and re-checked (no raw Markdown, no near-empty pages except one section-lead page).
14	draw.io files had unconnected arrows.	In the use-case and component `.drawio` files, arrows whose ends touch a shape are now attached to it (verified by re-rendering in draw.io). Sequence diagrams keep free-standing message arrows.